Digital payment adoption across Indian states, 2018–2026: building the state panel from PhonePe Pulse

BTP study, Part 1

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Abstract

This study builds a state-quarter panel of digital payment adoption in India. NPCI publishes UPI data by state only from April 2025. Thus the panel uses PhonePe Pulse, the open data of a UPI payment app, at one fixed commit. The panel has 35 state units and 34 quarters, from 2018Q1 to 2026Q2. It gives the transaction count, the amount, the registered users and the count by type. The denominator is the quarterly population from a linear interpolation of the NCP 2020 projections. All checks between the files of the source pass, and all values are nominal. The national count increases from 0.43 billion transactions in 2018Q4 to 38.66 billion in 2026Q2, an annual rate of 82.3 percent. The average transaction value rises to Rs. 1,907 in 2020Q3 and falls to Rs. 1,177 in 2026Q2. From 2021Q2, the retail share of the count increases from 37.3 to 63.9 percent. Across the 30 units of the main sample, the count per person in 2026Q2 goes from 3.15 to 100.30 transactions per quarter. The Spearman rank correlation of the state values in 2018Q4 and 2026Q2 is 0.892. A variance decomposition divides the differences between units into reach and intensity. Reach gives the larger contribution in 2018Q4 (0.776), and intensity gives the larger contribution in 2026Q2 (0.614).

1 Introduction

1.1 The question and the data gap

This study is Part 1 of a B.Tech project on digital payments and financial inclusion across the states of India. The project asks whether the states that adopted digital payments faster also became more financially inclusive and grew faster. It also asks in which direction the relation goes. Adoption can increase inclusion, inclusion can make adoption faster, or both can occur. The measure of adoption is the use of the Unified Payments Interface (UPI). UPI is the real-time payment system that the National Payments Corporation of India (NPCI) operates.

These questions need data on adoption for each state over many periods. Data for one period show which states have more adoption. They cannot show whether a change in adoption comes before or after a change in inclusion. The methods of the project use the changes within each state over time: fixed-effects panel regressions and tests of the direction of the relation. These methods need a state panel, that is, one value for each state in each period. The states also differ in population, so the panel must divide each total by the population of the state.

NPCI publishes UPI statistics, but its data by state start only in April 2025 [5, 1]. A panel from the NPCI data thus has no quarter before April 2025. It cannot show how adoption changed in the years before 2025. The project therefore needs another source with data by state over a longer period.

1.2 PhonePe Pulse as a proxy

PhonePe is a digital payment app that uses UPI. The company publishes PhonePe Pulse, an open data repository with transaction counts and amounts, and user counts, for each state and each quarter from 2018Q1 [6]. This study uses PhonePe Pulse as the measure of adoption. The data cover the payments of one app, not all UPI payments. Part 2 of the project compares the PhonePe values with the NPCI state data for the quarters that both sources cover.

Reddy et al. [7] used the state and district data of PhonePe Pulse to study how UPI spread across India. They found that the regions with high early adoption stayed ahead, with little change in the ranks of states and districts. Their paper uses an earlier release of the data. The release that this study uses restates all quarters with new definitions (Section 2.1). Thus their numbers and the numbers in this study are not comparable.

1.3 Aim and contents of this part

The aim of this part is to build the state-quarter panel of PhonePe adoption and its population denominator, to check it, and to describe it. The panel has 35 state units and 34 quarters, from 2018Q1 to 2026Q2. Statistics across states use a main sample of 30 units. All values in this part are nominal: the study does not adjust the amounts for changes in prices.

Section 2 describes the data sources and the files that the study uses. Section 3 explains how the raw files become the panel: the state units, the parsing rules, the data checks, the quarterly population and the measures. Section 4 describes adoption in India as a whole. Section 5 compares the states. Section 6 gives the limitations of the data, and Section 7 gives the conclusions. Appendix A lists the state units, and Appendix B gives the commands that make all the numbers again. Part 2 of the project converts the amounts to real values with consumer price indices. It also compares PhonePe Pulse with the NPCI state data and shows the data on maps. Part 3 tests whether the states with low adoption move closer to the leading states over time (convergence).

2 Data sources

The panel has data from PhonePe Pulse for adoption and from the population projections of the National Commission on Population (NCP 2020) for the denominator. This section describes each source and the files that the study uses.

2.1 PhonePe Pulse

PhonePe Pulse is a public Git repository on GitHub. In it, PhonePe publishes aggregate and anonymised counts of the transactions and users of its app [6]. The licence is the Community Data License Agreement – Permissive, Version 2.0 (CDLA-Permissive-2.0). This licence lets any person use, change and share the data, if the licence text goes with the shared data. PhonePe adds new data to the repository each quarter. Thus the study uses the data at one fixed commit: `257bf38`, of 24 September 2026. All PhonePe values in this study come from this commit.

The `data` folder of the commit has the sections `aggregated`, `map` and `top`. Each section has folders for transactions, users and merchants. The `aggregated` files give totals for India and for each state, with the transactions split by type. The `map` files give values by state (the "hover" files) and by district. The `top` files list the states and districts with the largest values. Each JSON file holds one quarter. For example, the file `2021/1.json` has the data for January to March 2021.

The commit is a new release of the data. Its README says that PhonePe restated the earlier quarters, from January–March 2018, with new definitions of the metrics. The README also advises users not to compare figures from this release with figures that PhonePe published

Table 1: PhonePe Pulse files used in the study. Unit: state-quarter. Sample: 36 raw state and union-territory names, 34 quarters, 2018Q1–2026Q2. Source: PhonePe [6], pinned commit.

Folder (under `data/`)	Content and use	Items
`map/transaction/hover`	Transaction count and amount by state	34 quarter files; 3,570 rows in the interim file
`map/user/hover`	Registered users by state	34 quarter files; in the same interim file
`aggregated/transaction/country/india/state`	Transaction count by type: P2P, Retail, Utility	36 names × 34 quarters; 3,570 rows in the interim file
`aggregated/transaction/country` and `aggregated/user/country`	National totals, used for checks only	34 quarter files each

before. It advises users not to join the data of this release and of earlier releases into one series. This study follows that rule: it uses only the files of the pinned commit.

Table 1 lists the files that the study uses, with their content and the number of items.

The transaction hover file of a quarter has one entry for each state and union territory. Each entry gives the number of transactions in the quarter (the count) and their total value (the amount). The README does not state the unit of the amount. This study reads it as nominal rupees. The user hover file gives the number of registered users of each state in the quarter. The README does not say whether this number is cumulative. The number never falls from one quarter to the next in any state unit (Section 3.3). This result agrees with a cumulative count of registrations. Each hover file has 36 raw names. The interim file `phonepe.csv` has 3,570 rows: one row for each state id, quarter and variable.

The aggregated state files split the count of each state by type. The README defines the types as follows. P2P transactions are person-to-person money transfers. Utility transactions are recharges and bill payments. Business transactions are payments to merchants. This type replaces the earlier types for financial services, merchants and others. The files use the label "Retail" for the business type, and this study uses the file labels. The README says that the types do not overlap and that they sum to the total count. The type files give counts only, with no amounts. Thus the study can split the count by type, but not the amount. The interim file `phonepe_types.csv` has 3,570 rows: one row for each state id, quarter and type. The national files of the `aggregated` section give the total count and users of India. The study uses them only to check the state files (Section 3.3).

2.2 Files not used

The study does not use the merchant files. The README defines registered merchants as the life-till-date count of unique merchants on the platform. In each quarter from 2018Q1 to 2019Q2, one or more raw units have no merchant file. A panel of merchants would need values for these gaps, and the data do not give them.

The study does not use the district files or the top lists. The unit of the project is the state. District names change over the period, so district data would need a separate crosswalk of district names. This study does not make one. The top lists give only the units with the largest values in a quarter, so they do not cover all units.

The pinned commit has no files on insurance, on app opens or on the device brands of users. The README still describes the user data as "broken down by devices". But the user files at this commit give only the registered count.

2.3 NCP 2020 population projections

The denominator is the projected population from the report of the Technical Group on Population Projections. The National Commission on Population issued this report, and the cover of the report gives the date November 2019 [4]. The report projects the population of India and of each state and union territory for each year from 2011 to 2036. For the states except Goa, and for Delhi and the union territory of Jammu and Kashmir, the report uses the component method. This method projects fertility, mortality and migration from the 2011 Census and the Sample Registration System. The report applies it to the north-eastern states other than Assam as one group. For Ladakh, the report uses the difference between its projections for the former state and the union territory of Jammu and Kashmir. For Goa and the other union territories, it uses a mathematical method. This method fits exponential growth rates by least squares to the Census growth rates from 1981 to 2011.

The study uses Table 8 of the report. This table gives the projected total population, in thousands of persons, on 1 March of each year. The study uses the years 2018 to 2027. In each year, the table has 37 rows for the states and union territories and a row for India. A Python module reads the table from the PDF of the report into the file `data/manual/population_ncp2020.csv`. Section 3.4 explains how the annual values become quarterly values, and why the panel needs the year 2027.

2.4 Data flow

Figure 1 shows the path of the data from the raw files to the outputs of the study. The top row has the raw sources. The second row has the Python modules that read them. Each module writes interim files in a long format, with one row for each state id, period and variable. The module `parse_phonepe_types` also writes the file of data checks. The R script `part1_study.R` reads the interim files. It writes all the figures, the tables and the numbers of this study. The text gets each number from the file `numbers.tex` that the script writes, so the text has no hand-typed data value (Appendix B).

3 Construction of the panel

This section explains how the raw files become the panel. It describes the state units, the parsing rules, the data checks, the quarterly population and the measures.

3.1 State units and the crosswalk

The sources do not use the same names for the states. For example, NCP 2020 writes "NCT of Delhi" and PhonePe Pulse writes "delhi". A crosswalk file gives the state id for each name variant. Before the lookup, a function changes the name to lower case, replaces "&" with "and" and removes extra spaces. The crosswalk has 46 name variants and 35 state ids. It also has old names, for example "Orissa" for Odisha. If a source has a name with no row in the crosswalk, the parse stops. Thus no state can drop out of the panel without an error.

The sources do not divide the country in the same way. PhonePe Pulse gives Jammu and Kashmir and Ladakh as separate units, and NCP 2020 does the same. NCP 2020 gives Dadra and Nagar Haveli and Daman and Diu in separate rows, but PhonePe Pulse gives them as one unit. A state id must cover the same area in all sources. Thus the crosswalk puts Ladakh into the id of Jammu and Kashmir. It also puts Dadra and Nagar Haveli and Daman and Diu into one id. Section 3.2 gives the rule that sums the parts of these merged ids.

The panel has 35 state ids. This study calls the area of each id a state unit, or a unit. The main sample has 30 units: the 28 states, Delhi, and Jammu and Kashmir with Ladakh. The other 5 units are the union territories of Andaman and Nicobar Islands, Chandigarh, Dadra

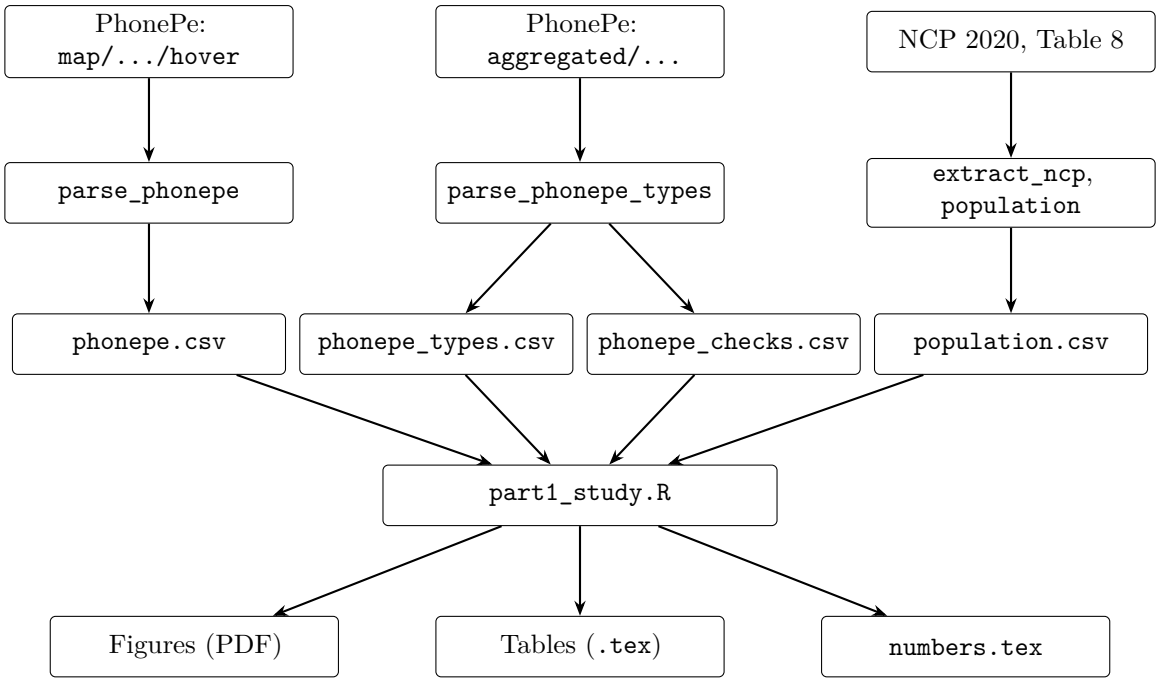


Figure 1: Data flow from the raw sources to the study outputs. Raw files: PhonePe [6] and National Commission on Population [4]. Top row: raw sources; second row: Python modules; third row: interim files; then the R script and its outputs. All interim files except the check file have the columns `state_id`, `period`, `variable`, `value` and `source`.

and Nagar Haveli and Daman and Diu, Lakshadweep, and Puducherry. The figures and tables show all 35 ids and mark these 5 union territories. Statistics across units, such as variances, rank correlations and the variance decomposition, use only the main sample. This sample rule is the same in all parts of the project. Shares of the national total use all 35 ids, because the national total is their sum. Appendix A lists the units with their raw names in each source. The two-letter codes in the figures follow ISO 3166-2:IN [8].

3.2 Parsing and the summation rule

Python modules read the JSON files of each quarter. The PhonePe module takes the state name, the count and the amount from the transaction hover file. It takes the registered users from the user hover file. The type module reads the aggregated file of each state folder. A folder name uses hyphens in place of spaces, for example `jammu-&-kashmir`. The module changes each hyphen to a space and then finds the state id in the crosswalk. The modules do not round the values.

A merged id gets the sum of its parts in each quarter. The rule applies to all additive variables: count, amount, users, type counts and population. Before the sum, the rule finds all the parts that a source gives for a merged id over the full period. It then checks that each quarter has all these parts. If a part is absent in a quarter, the parse stops. This check prevents a merged id from losing a part in some quarters and not in others.

The type module also compares values that the repository gives in more than one file. For each state and quarter, the sum of the type counts must equal the count in the hover file. For each quarter, the national count file must equal the sum of the state counts. This file gives the type counts of India in the `aggregated` section. The national user file must equal the sum of the state user counts. If one of these comparisons fails, the module stops and names the file.

A type file can lack the entry for a type. The module then tests the present types. If they sum to the hover count of that state and quarter, the module sets the absent type to zero. If

Table 2: Data checks. Unit: the item named in each row (state-quarters, state-quarter-type entries, quarters, pairs of consecutive quarters, state rows or panel cells). Sample: the 36 raw PhonePe units for the type and national checks, the 35 state ids for the user and panel checks, and the state rows of NCP 2020, Table 8; 34 quarters, 2018Q1–2026Q2. The last column is the largest absolute difference for the PhonePe rows and the ratio of the state sum to the India row for the NCP rows. Source: PhonePe [6], National Commission on Population [4].

Check	Items	Flagged	Difference or ratio
PhonePe: sum of the three type counts against the state count (state-quarters)	1,224	0	0
PhonePe: absent type entries set to zero (state-quarter-type entries). Flagged: Lakshadweep, 2018Q1, Utility	3,672	1	0
PhonePe: national count file against the sum of the state counts (quarters)	34	0	0
PhonePe: national user file against the sum of the state user counts (quarters)	34	0	0
PhonePe: registered users lower than in the previous quarter (pairs of consecutive quarters)	1,155	0	–
NCP 2020, 2018: sum of the state rows divided by the India row (state rows)	37	–	1.0000015
NCP 2020, 2027: sum of the state rows divided by the India row (state rows)	37	–	0.9999993
Population: missing cells in the panel (35 ids × 34 quarters)	1,190	0	–

they do not, the module stops and names the file. Thus a zero in the panel comes from the data, not from an assumption. The type files have 1 absent entry: the file of Lakshadweep for 2018Q1 has no Utility entry. Its P2P and Retail counts sum to the hover count of Lakshadweep for that quarter. Thus the module sets the Utility count to zero.

3.3 Data checks

Table 2 gives the results of all the data checks. Each row names the check and the unit of its items. The column "Flagged" gives the number of items that fail the check or, for the zero-fill row, the number of absent entries.

The PhonePe checks pass with no exception. In all 1,224 state-quarters, the type counts sum to the hover count. The type checks use the 36 raw units before the merge, so Ladakh and Jammu and Kashmir count as separate units. Of the 3,672 type entries, 1 is absent, and the module sets it to zero (Section 3.2). The national count file and the national user file equal the sums of the state files in all 34 quarters. Thus the state files account for the full national totals, and the panel can form national totals as the sum of the state ids.

The registered users of a state unit never fall from one quarter to the next. The check covers 1,155 pairs of consecutive quarters, and 0 pairs show a fall. A count of active users can fall from one quarter to the next, but a cumulative count cannot. The result thus agrees with a cumulative count of registrations. It does not prove that the count is cumulative.

The NCP rows check the population table. The sum of the 37 state rows divided by the India row is 1.0000015 in 2018 and 0.9999993 in 2027. The sum of the state rows minus the India row is 2 thousand persons in 2018 and −1 thousand persons in 2027. Table 8 rounds each row to the nearest thousand, so the sum of the rows can differ from the India row. With 37 state rows and one India row, this rounding can give a difference of up to 19 thousand persons. The two differences are smaller than this value. The last row shows that the population panel has a value for every state id and quarter.

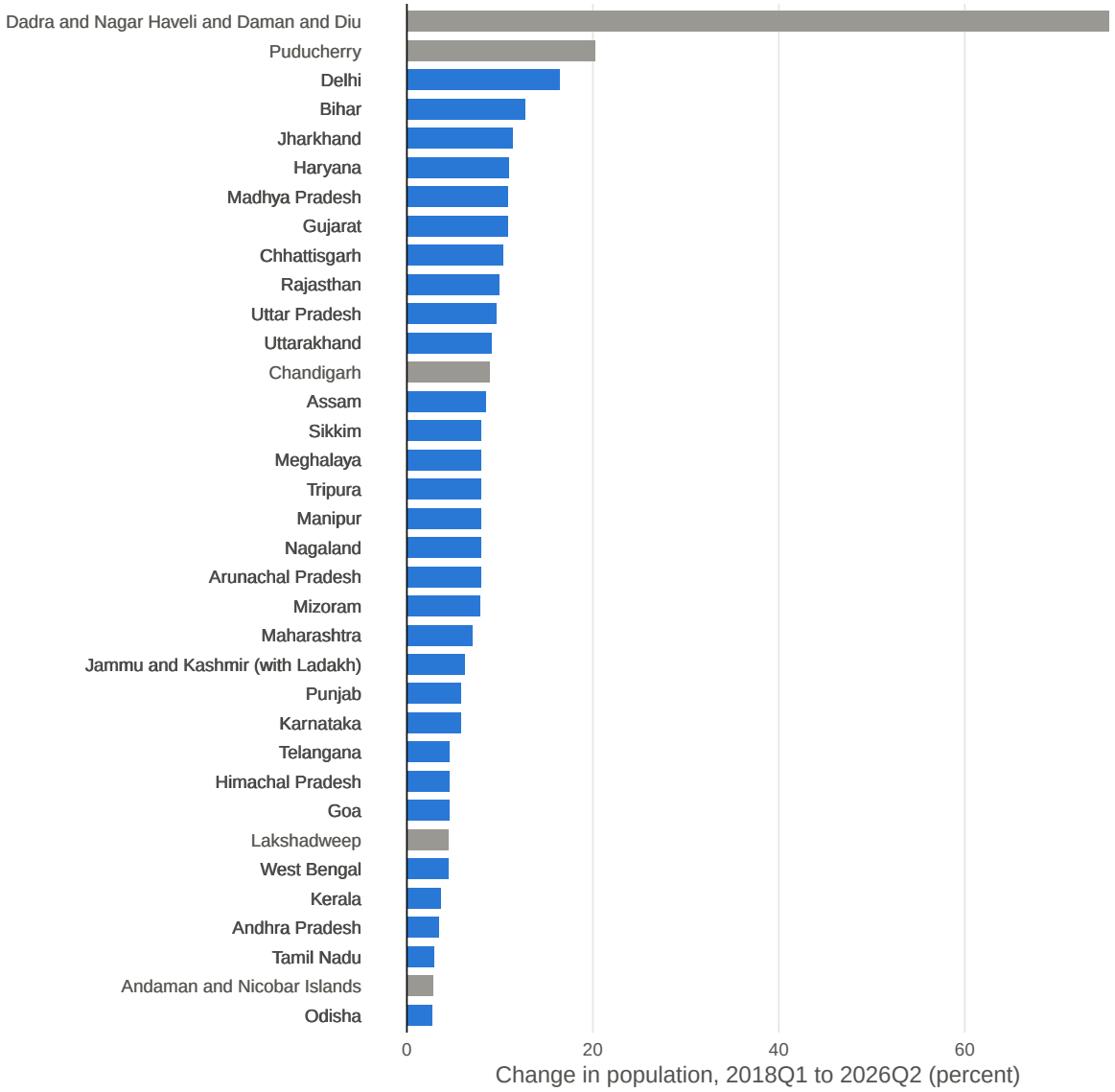


Figure 2: Population growth by state unit, 2018Q1 to 2026Q2. Unit: percent change. Sample: all 35 state units; the other union territories are marked. Source: interpolation of the NCP 2020 projections [4].

3.4 Population by quarter

NCP 2020 gives the population on 1 March of each year, but the panel needs one value for each quarter. The study uses the 1 March value of year y as the value of the first quarter of y. For each later quarter, it adds one quarter of the change to the next year:

$$P_{y,q} = P_y + \frac{q-1}{4}\,(P_{y+1} - P_y), \qquad q = 1, \ldots, 4. \tag{1}$$

Here P_y is the population on 1 March of year y, and $P_{y,q}$ is the population in quarter q of year y. This is a linear interpolation between the 1 March values. It assumes an even change within each year. The quarters of 2026 after the first need the value for 2027, so the study uses Table 8 up to 2027.

The 35 state ids have a total population of 1,318.7 million in 2018Q1 and 1,426.0 million in 2026Q2. This is an increase of 8.1 percent. The growth differs between the units. Figure 2 shows the change in population of each state id from 2018Q1 to 2026Q2.

The figure sorts the state ids by growth, and grey bars mark the union territories outside the main sample. The largest growth is 75.5 percent, for Dadra and Nagar Haveli and Daman and

Diu. The data of this study do not show the reason for this large value. A larger population gives smaller values per person, so the per-person values of this unit depend on this projection. In the main sample, the largest growth is 16.5 percent, for Delhi. The smallest growth is 2.8 percent, for Odisha. Bihar, Jharkhand and Haryana are near the top of the main sample, and Tamil Nadu, Andhra Pradesh and Kerala are near the bottom. In a unit with fast population growth, a per-person value grows more slowly than its total. Thus these differences in growth matter for the comparison of units over time.

3.5 Measures

For state s and quarter t, let C_{st} be the transaction count, A_{st} the amount in nominal rupees, U_{st} the registered users and P_{st} the population. Let C^k_{st} be the count of type k, where k is P2P, Retail or Utility. The study uses these measures (without the subscripts):

$$\begin{aligned} \text{count per person} &= C/P, & \text{amount per person} &= A/P, \\ \text{users per 1000 persons (reach)} &= 1000\,U/P, & \text{transactions per user (intensity)} &= C/U, \\ \text{average transaction value} &= A/C, & \text{type share} &= 100\,C^k/C. \end{aligned}$$

The count per person is the main measure of adoption. Its unit is transactions per person per quarter. It gives the number of PhonePe transactions in a quarter for each resident of the state. The amount per person is in rupees per person per quarter. It gives the value of these transactions for each resident. Reach is in registered users per 1000 persons. It shows how many registered users the app has for each 1000 persons in a state. It does not show how many of them used the app in the quarter. Intensity is in transactions per registered user per quarter. It shows how often a registered user pays with the app. The average transaction value is in rupees per transaction. The type share is the percent of the count of a state that is of type k. The P2P, Retail and Utility shares of a state add up to the whole count.

The growth of a variable X from quarter $t-1$ to quarter t is $100\,(X_t/X_{t-1}-1)$ percent. The annual rate of growth from a first to a last quarter is $100\,[(X_1/X_0)^{1/n}-1]$ percent. Here X_0 and X_1 are the values in the first and the last quarter, and n is the number of years between them. The study gives growth rates from 2018Q4, not from 2018Q1. The national count is 0.14 billion in 2018Q1 and 0.43 billion in 2018Q4, so rates from 2018Q1 start from a much smaller base. Also, the national retail share moves between 38.4 and 64.2 percent during 2018.

Count per person is the product of reach and intensity: $C/P = (U/P)\,(C/U)$. Let $y = \ln(C/P)$, $r = \ln(U/P)$ and $i = \ln(C/U)$. Then $y = r + i$ for each state. Across the units of the main sample in one quarter, the variance of y is thus $\text{var}(y) = \text{cov}(r, y) + \text{cov}(i, y)$. Division by $\text{var}(y)$ gives

$$1 = \frac{\text{cov}(r, y)}{\text{var}(y)} + \frac{\text{cov}(i, y)}{\text{var}(y)}. \tag{2}$$

The first term is the contribution of reach, and the second term is the contribution of intensity. Each term is the slope of a least-squares line of r (or i) on y across the units. The reach term gives the part of the differences in adoption between units that goes with differences in reach. The intensity term gives the part that goes with differences in use per user. A term can be negative, and the other term is then larger than 1. The decomposition shows how the differences divide between reach and intensity. It does not show a cause. Section 5 gives the results for each quarter.

4 Adoption in India as a whole

This section describes the national series. The national value of a variable is the sum over the 35 state ids. Section 3.3 shows that this sum equals the national files of PhonePe Pulse in each

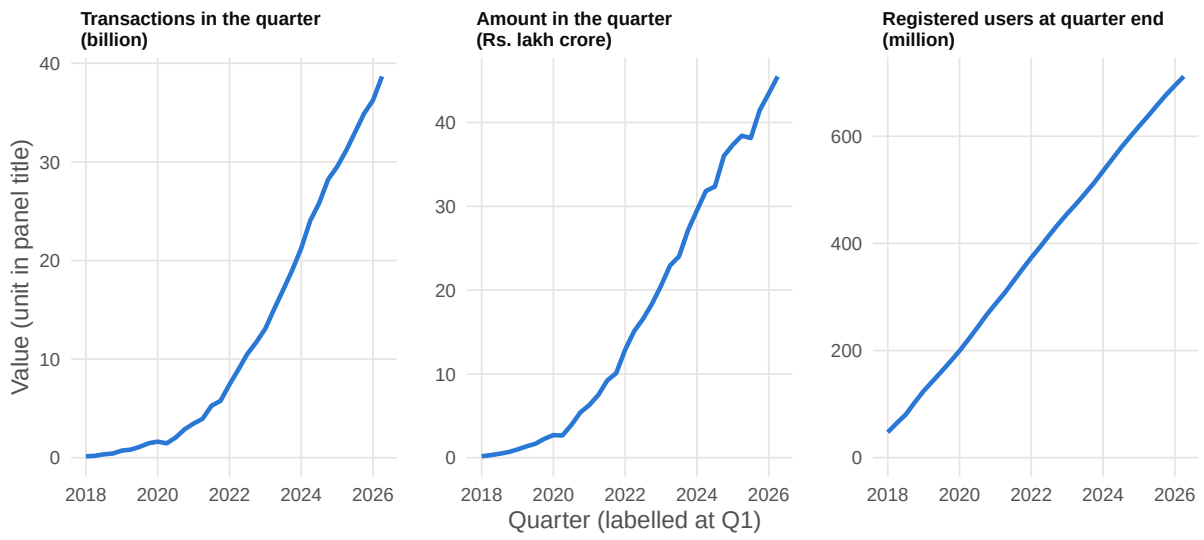


Figure 3: National transaction count (billion), transaction amount (Rs. lakh crore) and registered users (million), by quarter. Unit: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [6].

quarter. All amounts are nominal. Section 4.1 gives the totals and their growth (Figure 3 and Table 3). Section 4.2 gives the growth from quarter to quarter (Figure 4). Section 4.3 gives the average transaction value (Figure 5). Section 4.4 gives the shares of the transaction types (Figure 6).

4.1 Totals and growth

Figure 3 shows the three national totals in each quarter: the transaction count, the amount and the registered users. Each panel has its own vertical scale, and each scale starts at zero.

The count increases from 0.14 billion transactions in 2018Q1 to 38.66 billion in 2026Q2. Thus the count of 2026Q2 is 276 times the count of 2018Q1. The amount increases from Rs. 0.17 lakh crore to Rs. 45.50 lakh crore, a ratio of 264. The registered users increase from 47.5 million to 711.7 million, a ratio of 15.

These ratios start from the small values of 2018Q1. Section 3.5 gives the reasons to measure growth from 2018Q4. From 2018Q4 to 2026Q2, the annual rate of growth is 82.3 percent for the count and 75.4 percent for the amount. The annual rate for the registered users is 29.4 percent. The count grows faster than the amount, so the average value of a transaction decreases between the two quarters (Section 4.3). The users grow more slowly than the count, so the number of transactions per user increases.

The shape of the user curve differs from the shapes of the other two curves. The user curve is almost a straight line. Each full year from 2019 to 2025 adds between 76.9 and 86.3 million registered users. An increase of almost the same size each year gives a rate of growth that decreases over time. The count and the amount curves rise slowly up to 2020 and more steeply after 2021.

Per person, the count increases from 0.11 transactions per person per quarter in 2018Q1 to 0.32 in 2018Q4. In 2026Q2, it is 27.11. The registered users per 1000 persons increase from 36 in 2018Q1 to 78 in 2018Q4 and 499 in 2026Q2. This last value is not the share of persons who use PhonePe. The check in Section 3.3 agrees with a cumulative count of registrations. Such a count can include registrations that are no longer in use. The data also do not show whether one person can have more than one registration.

Table 3 gives the national values for each calendar year. The count and the amount are sums over the quarters of the year. The users are the value at the end of the year. The count per

Table 3: National annual totals and growth. Unit: count in billion, amount in Rs. lakh crore (nominal), users at year end in million, count per person per year, average value in Rs., growth of count in percent. Sample: all 35 state units. Period: calendar years 2018–2026 (2026 has two quarters). Source: PhonePe [6]; population from National Commission on Population [4].

Year	Count (billion)	Amount (Rs. lakh crore)	Users at year end (million)	Count per person	Average value (Rs.)	Growth of count (percent)
2018	1.11	1.63	103.0	0.8	1,467	–
2019	4.12	6.29	179.9	3.1	1,525	271.9
2020	8.02	14.65	266.2	5.9	1,827	94.5
2021	18.45	33.09	351.7	13.5	1,793	130.0
2022	38.64	62.97	436.0	28.0	1,630	109.4
2023	64.30	94.60	513.0	46.2	1,471	66.4
2024	99.33	129.71	598.7	70.8	1,306	54.5
2025	128.54	155.38	676.8	90.8	1,209	29.4
2026 (Q1–Q2)	74.92	88.94	711.7	52.6	1,187	–

person is the annual count divided by the mean population of the quarters of the year. The year 2026 has only two quarters, so the table gives no growth rate for it.

The annual count increases in each full year. Its rate of growth is 271.9 percent in 2019, 94.5 percent in 2020 and 130.0 percent in 2021. The rate of 2020 is thus lower than the rates of the years before and after it. The only fall of the national count, in 2020Q2, is in this year (Section 4.2). After 2021, the rate decreases in each year. It is 109.4 percent in 2022, 66.4 percent in 2023, 54.5 percent in 2024 and 29.4 percent in 2025.

The count per person increases from 0.8 transactions per person in 2018 to 90.8 in 2025. In 2025, the app records 128.54 billion transactions with a total value of Rs. 155.38 lakh crore. The annual average value is largest in 2020, at Rs. 1,827. It decreases in each later year, and it is Rs. 1,209 in 2025.

4.2 Quarter-on-quarter growth

Figure 4 shows the change of the national count and of the national amount from the previous quarter, in percent. A bar below zero shows a quarter with a fall.

The growth of the count is largest in the first quarters, when the base is small. The largest value is 79.2 percent, in 2018Q3. From 2018Q2 to 2024Q2, the median growth of the count is 17.7 percent per quarter. In the last eight quarters, from 2024Q3 to 2026Q2, the median is 5.9 percent per quarter. Thus the count still grows in each recent quarter, but at a lower rate.

The only quarter with a fall in the national count is 2020Q2, the quarter from April to June 2020. The change of the count in this quarter is −10.6 percent. The count goes from 1.63 billion in 2020Q1 to 1.46 billion in 2020Q2. The change of the amount is −1.9 percent, so the amount falls less than the count. The count falls in 16 of the 35 state ids. In the other ids, the count of 2020Q2 is equal to or higher than the count of 2020Q1.

The fall does not continue. In 2020Q3, the count grows by 40.3 percent to 2.05 billion, which is more than the count of 2020Q1. The Ministry of Home Affairs ordered a national lockdown from 25 March 2020 [2]. Thus 2020Q2 is the first full quarter after the start of the lockdown. The data of this study do not test whether the lockdown caused the fall.

The amount falls in the quarters 2020Q2 and 2025Q3. The fall in the later quarter is small: the change is −0.7 percent. The count does not fall in that quarter, so the average value of a transaction decreases.

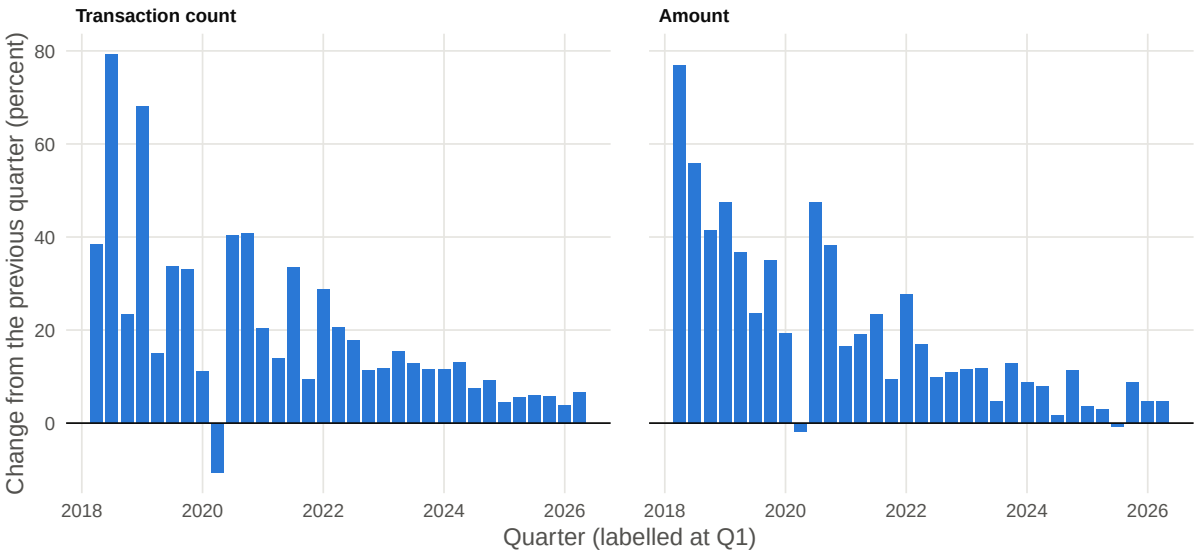


Figure 4: Quarter-on-quarter growth of the national transaction count and amount. Unit: percent change from the previous quarter. Sample: national total of all 35 state units. Period: 2018Q2–2026Q2. Source: PhonePe [6].

4.3 Average transaction value

The average transaction value is the amount divided by the count. Figure 5 shows it for India and for the units of the main sample. The national line is the national amount divided by the national count. The state line is the median of the 30 state values in each quarter. The band goes from the first to the third quartile of the state values.

The national value is Rs. 1,232 in 2018Q1. It rises, with large changes from quarter to quarter, to a peak of Rs. 1,907 in 2020Q3. After the peak, it is lower than in the previous quarter in 17 of 23 quarters. In 2026Q2, it is Rs. 1,177, which is lower than the value of 2018Q1. In 2020Q2, the count falls more than the amount (Section 4.2), so the average value rises in that quarter.

The national value is a mean of the state values with the count of each unit as its weight. The state median gives each unit the same weight. The two lines thus differ, but they have a similar shape. The median has its largest value, Rs. 1,949, in 2020Q4. In 2026Q2, the median is Rs. 1,154.

The width of the band shows how much the units differ. The interquartile range is Rs. 532 in 2018Q4. It has its largest value, Rs. 628, in 2021Q1. After that quarter, it decreases with some rises, and it is Rs. 156 in 2026Q2. Thus the average values of the units are closer to each other in 2026Q2 than in 2018Q4. In 2026Q2, the highest value in the main sample is Rs. 1,623, in Manipur. The lowest is Rs. 946, in Assam (Table 4).

The fall of the average value and the rise of the retail share occur in the same years. From 2021Q2, the retail share rises in most quarters (Section 4.4), and the average value falls. This pattern agrees with a larger number of small payments to merchants. The study does not test this explanation. The type files give counts and no amounts, so the study cannot compute the average value of each type. The values are also nominal. Thus the study does not separate a change in prices from a change in the size of payments.

4.4 Transaction types

Figure 6 shows the national shares of the three types in the count: P2P, Retail and Utility. In each quarter, the three shares add up to the whole count.

The shares change by large amounts in 2018. In 2018Q1, the retail share is 64.2 percent

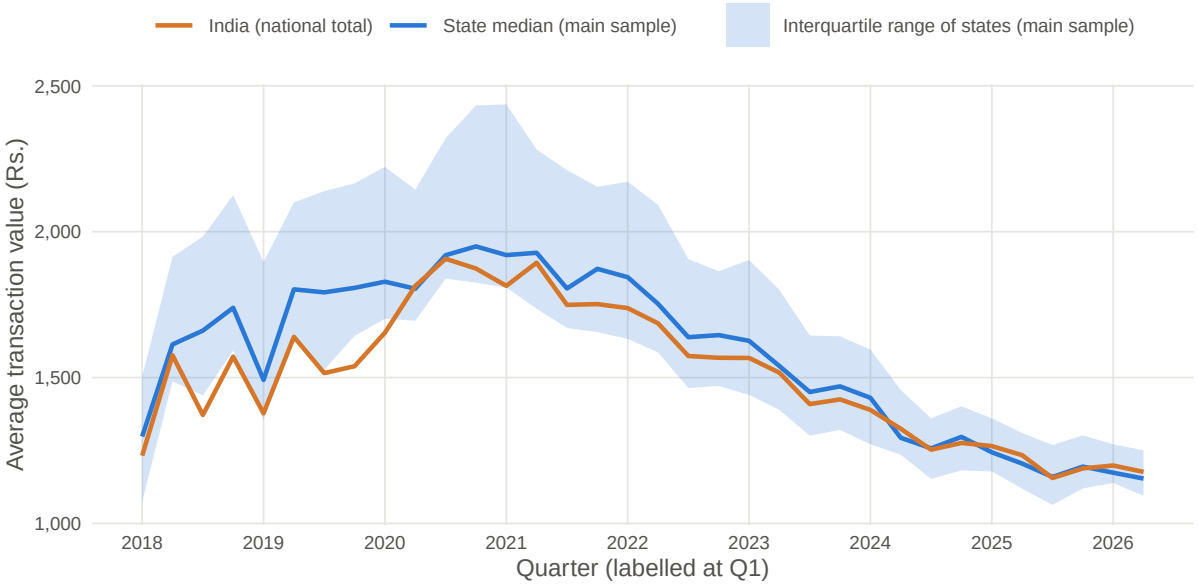


Figure 5: Average transaction value. Unit: Rs. per transaction (nominal), amount divided by count. Sample: national value, and median and interquartile range across the 30 units of the main sample. Period: 2018Q1–2026Q2. Source: PhonePe [6].

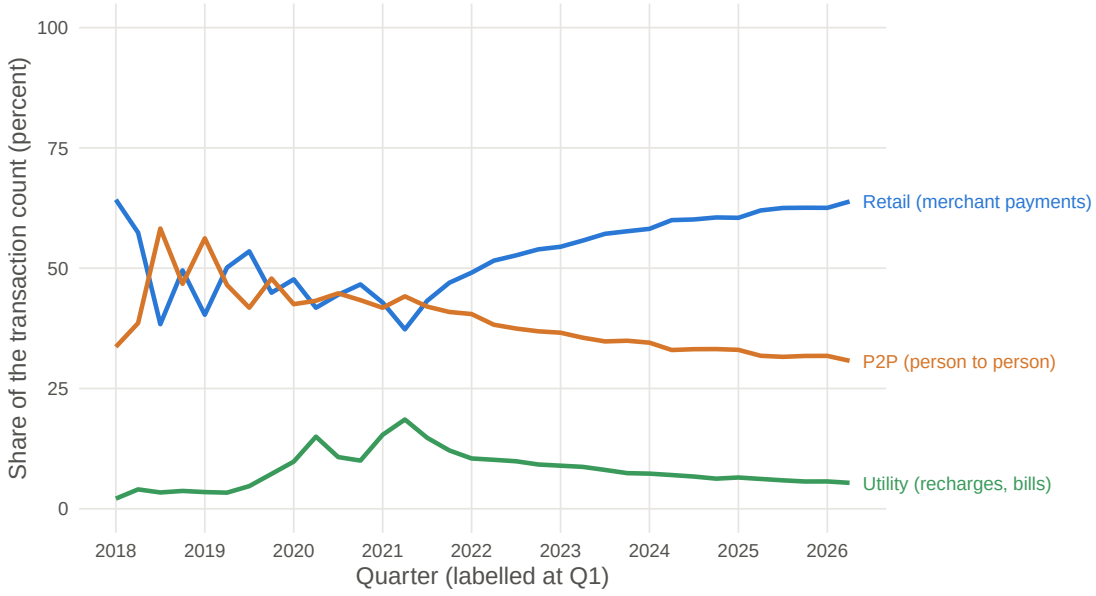


Figure 6: National shares of the transaction count by type: P2P, Retail and Utility. Unit: percent of the national count. Sample: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [6].

and the P2P share is 33.6 percent. During 2018, the retail share moves between 38.4 and 64.2 percent. In 2018Q4, the retail share is 49.5 percent, the P2P share is 46.7 percent and the utility share is 3.7 percent. The data do not show the reason for these changes (Section 6).

In 2019Q1–2019Q3, the retail and P2P shares differ by up to 15.9 percentage points, in 2019Q1. From 2019Q4 to 2021Q3, the two shares differ by 6.9 percentage points or less. In this window, their order changes more than once. The utility share also has its largest value in this window: 18.6 percent, in 2021Q2. In the same quarter, the retail share has its lowest value, 37.3 percent.

From 2021Q3, the retail share is above the P2P share in every quarter. The retail share rises to 63.9 percent in 2026Q2, and the P2P share falls to 30.8 percent. The utility share falls from its peak to 5.4 percent. Thus, in the recent quarters, payments to merchants are the largest type in the count. The retail share of 2026Q2 is close to that of 2018Q1, but the count is 276 times larger.

The type shares are shares of the count, not of the amount. A payment to a merchant and a transfer to a person each count as one transaction. The data do not show which type has the larger payments. Thus the type shares do not show how the value of all payments divides between the types.

5 Adoption across states

This section compares the state units. Statistics across units use the 30 units of the main sample. The figures and Table 4 also show the 5 other union territories. The reference quarters are 2018Q4 and 2026Q2. Section 5.1 compares the count per person (Figure 7 and Table 4). Section 5.2 measures the concentration of activity in a few units (Table 5). Section 5.3 measures how stable the order of the units is. Sections 5.4 and 5.5 divide the count per person into reach and intensity (Figures 8 and 9, Table 6). Section 5.6 compares the retail share with the count per person (Figure 10).

5.1 Transactions per person

Figure 7 shows the count per person of each state unit in 2018Q4 and in 2026Q2. The horizontal axis has a log scale, so equal distances show equal ratios. The open circle gives the value of 2018Q4, and the filled circle gives the value of 2026Q2. The units are in the order of their values in 2026Q2. Grey names mark the other union territories.

In 2018Q4, the count per person in the main sample goes from 0.06 transactions per person per quarter, in Mizoram, to 1.26, in Delhi. The ratio of the highest to the lowest value is 22.6. This ratio uses the unrounded values, so it differs from the ratio of the two rounded values in the text. The median unit has 0.22 transactions per person per quarter. In this quarter, 10 units have a value above the national value of 0.32.

In 2026Q2, the lowest value is 3.15, in Manipur, and the highest value is 100.30, in Telangana. The ratio of the highest to the lowest value, also from unrounded values, is 31.8. The median unit has 18.78 transactions per person per quarter. In this quarter, 9 units are above the national value of 27.11. The national value is a mean of the state values with population weights, so it is not the value of a typical unit.

Every unit of the main sample has a higher value in 2026Q2 than in 2018Q4. From 2018Q4 to 2026Q2, the annual rate of growth of the count per person goes from 60.2 percent, in Manipur, to 98.4 percent, in Arunachal Pradesh. Part 3 of the project tests whether the units with low values grow faster than the units with high values. This study does not test this relation.

Table 4 gives the values of all state units in 2026Q2, in the order of the count per person. Next to the count per person, it gives the population, the amount per person, reach, intensity,

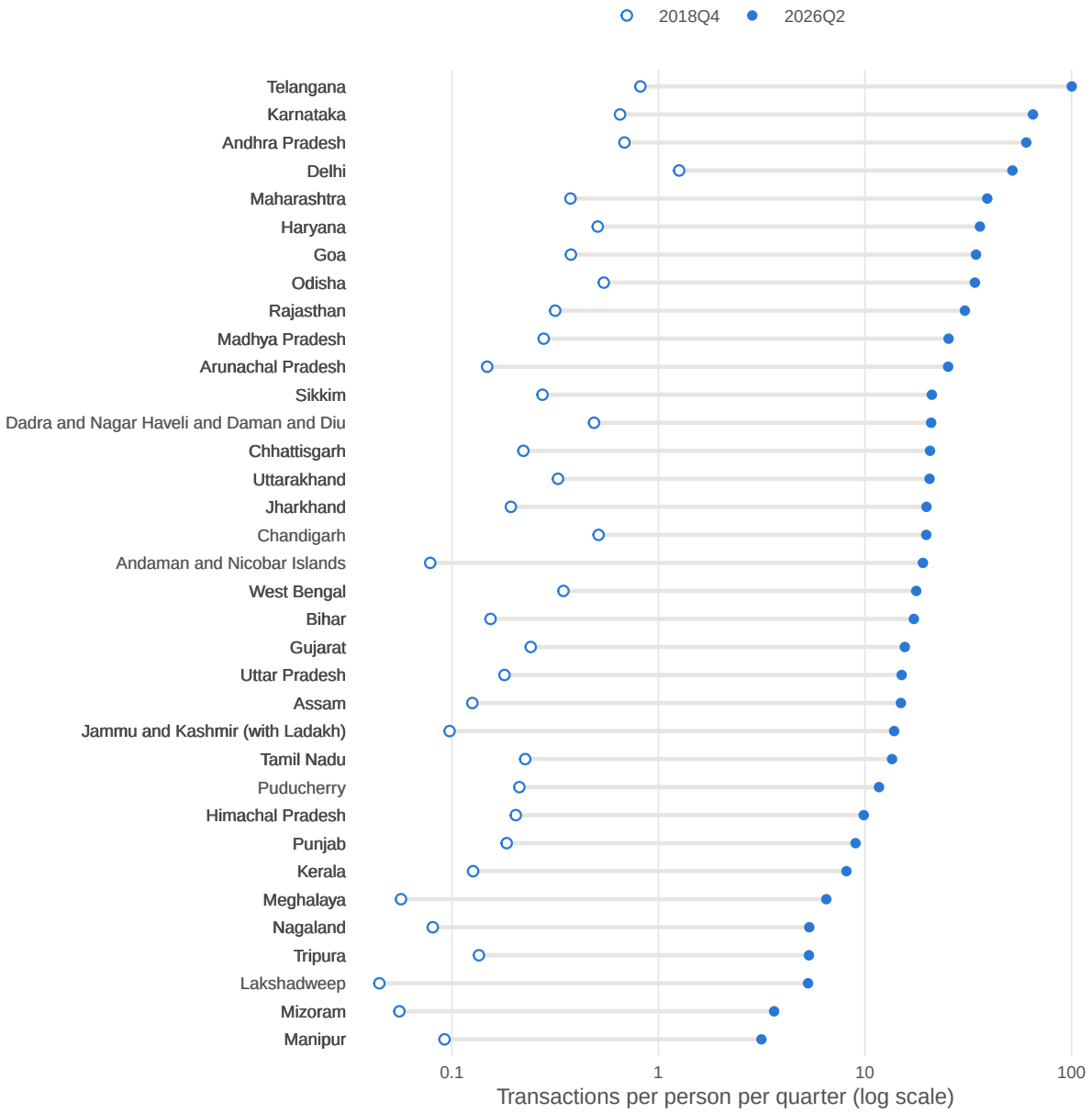


Figure 7: Transaction count per person per quarter, by state unit, in 2018Q4 and 2026Q2. Unit: transactions per person per quarter. Sample: all 35 state units; the other union territories are marked. Source: PhonePe [6]; population from National Commission on Population [4].

the average value and the retail share. Thus the table shows the measures of Section 3.5 side by side for each unit.

The five units with the highest count per person are Telangana, Karnataka, Andhra Pradesh, Delhi and Maharashtra. The five units with the lowest values, from the lowest, are Manipur, Mizoram, Tripura, Nagaland and Meghalaya. All five are in the north-east of India. Each of these five units has lower reach and lower intensity than each unit of the top five.

The amount per person goes from Rs. 4,769 per quarter in Mizoram to Rs. 132,629 in Telangana. The ratio of the two unrounded values is 27.8. The amount per person is the count per person times the average value. The average values of the units differ much less than their counts per person, so the two ratios are of similar size.

The size of a unit does not decide its place in the table. The three units with the largest population are Uttar Pradesh, Bihar and Maharashtra. Of these units, 2 have a count per person below the median of the main sample. Thus a unit with a large population can have a low count per person.

5.2 Concentration

Table 5 shows how much of the national activity is in a few state units. For each reference quarter, it takes the five units with the largest transaction count. It gives their shares of the national count, amount, registered users and population. Its last row gives the smallest number of units that together have half of the national count. These statistics use all 35 ids, because the national total is their sum.

In 2018Q4, the five units with the largest count are Maharashtra, Karnataka, Uttar Pradesh, Andhra Pradesh and West Bengal. They have 46.2 percent of the count and 42.2 percent of the population. In 2026Q2, the five units are Maharashtra, Karnataka, Telangana, Uttar Pradesh and Andhra Pradesh. Their share of the count is 52.6 percent, and their share of the population is 37.5 percent. The set of five units is not the same in the two quarters. Their population share is also different. The share of the amount increases from 44.1 to 53.7 percent.

In both quarters, the five units have a larger share of the count than of the population. The difference is larger in 2026Q2. Their share of the registered users is 43.7 percent in 2018Q4 and 41.9 percent in 2026Q2. In 2026Q2, this share is closer to the population share than to the share of the count. Thus the registered users of these five units make more transactions per user than the users of the other units.

Half of the national count comes from 6 units in 2018Q4 and from 5 units in 2026Q2. Thus, at the end of the period, fewer units have half of the count, and the top five units have a larger share of it.

5.3 Rank persistence

The Spearman rank correlation of the count per person in 2018Q4 and in 2026Q2 is 0.892. It uses the 30 units of the main sample, and average ranks for equal values. A value of 1 means the same order in both quarters.

Of the 30 units, 19 move by 3 places or fewer between the two quarters (Figure 7). Thus most units move by few places. Arunachal Pradesh moves up the most, from rank 22 to rank 11. Tamil Nadu and West Bengal move down the most, by 7 places each. Delhi, the highest unit in 2018Q4, is still among the top five in 2026Q2.

This result agrees with Reddy et al. [7], who found little change in the ranks of states in an earlier release of the data. The two studies use different releases, so their numbers are not comparable (Section 1). A stable order does not mean that the differences in level stay the same. Part 3 of the project measures how these differences change over time.

Table 4: State values in 2026Q2. Unit: population in million; count per person per quarter; amount per person in Rs. per quarter (nominal); users per 1000 persons; transactions per user per quarter; average value in Rs.; retail share in percent. Sample: all 35 state units, sorted by count per person; † marks the 5 union territories outside the main sample. Source: PhonePe [6]; population from National Commission on Population [4].

Name	Population (million)	Count per person	Amount per person (Rs.)	Users per 1000 persons	Count per user	Average value (Rs.)	Retail share (percent)
Telangana	38.66	100.3	132,629	943	106.3	1,322	66.0
Karnataka	69.05	65.1	76,649	761	85.5	1,178	67.5
Andhra Pradesh	53.73	60.4	86,051	639	94.6	1,424	61.0
Delhi	22.64	51.8	51,967	939	55.1	1,004	68.5
Maharashtra	129.52	39.1	42,197	649	60.2	1,079	67.4
Haryana	31.38	36.0	40,263	728	49.5	1,117	67.9
Goa	1.60	34.5	43,161	842	41.0	1,251	67.9
Odisha	44.69	34.1	37,283	505	67.4	1,094	61.2
Rajasthan	83.82	30.5	38,052	521	58.4	1,249	64.2
Madhya Pradesh	89.89	25.4	28,572	446	57.0	1,125	63.8
Arunachal Pradesh	1.61	25.3	29,550	592	42.7	1,169	60.8
Sikkim	0.71	21.1	25,729	574	36.7	1,221	58.3
Dadra and Nagar Haveli and Daman and Diu†	1.58	20.9	16,808	434	48.2	804	69.3
Chhattisgarh	31.29	20.7	21,684	422	49.0	1,050	62.3
Uttarakhand	12.02	20.6	21,235	562	36.6	1,033	66.4
Jharkhand	41.07	19.9	22,132	384	51.6	1,115	61.2
Chandigarh†	1.27	19.8	19,877	653	30.3	1,004	71.0
Andaman and Nicobar Islands†	0.41	19.1	25,704	543	35.1	1,347	63.4
West Bengal	100.60	17.7	21,986	430	41.2	1,241	56.4
Bihar	132.70	17.2	20,181	338	51.1	1,170	56.6
Gujarat	74.28	15.6	17,471	486	32.1	1,121	65.4
Uttar Pradesh	243.31	15.1	16,188	372	40.5	1,075	62.6
Assam	36.79	14.9	14,126	383	39.0	946	57.3
Jammu and Kashmir (with Ladakh)	14.22	13.8	15,360	415	33.4	1,109	62.3
Tamil Nadu	77.57	13.5	16,974	519	26.1	1,255	63.6
Puducherry†	1.77	11.7	13,528	409	28.6	1,157	69.1
Himachal Pradesh	7.59	9.9	10,819	474	20.8	1,096	64.1
Punjab	31.36	9.0	11,788	450	20.0	1,309	62.7
Kerala	36.23	8.1	9,260	450	18.1	1,139	65.1
Meghalaya	3.45	6.5	7,059	238	27.3	1,086	48.4
Nagaland	2.30	5.4	7,123	262	20.5	1,325	49.2
Tripura	4.27	5.4	6,265	291	18.4	1,170	56.7
Lakshadweep†	0.07	5.3	5,294	263	20.2	998	67.0
Mizoram	1.27	3.6	4,769	274	13.3	1,313	48.2
Manipur	3.32	3.2	5,118	257	12.2	1,623	43.5

† Union territory outside the main sample.

Table 5: Concentration of activity in the five units with the largest transaction count. Unit: share of the national total, in percent. Sample: all 35 state units. Periods: 2018Q4 and 2026Q2. Source: PhonePe [6]; population from National Commission on Population [4].

Share of the top five (by count)	2018Q4	2026Q2
Transaction count (percent)	46.2	52.6
Amount (percent)	44.1	53.7
Registered users (percent)	43.7	41.9
Population (percent)	42.2	37.5
Units needed for half of all transactions	6	5

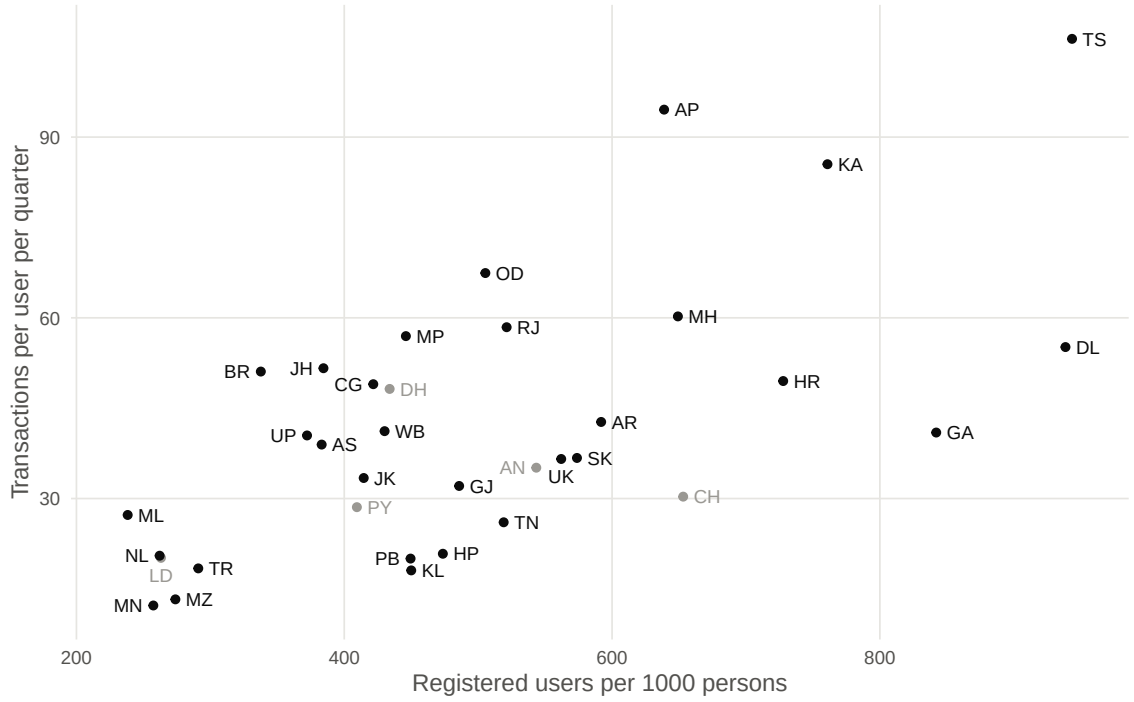


Figure 8: Reach against intensity in 2026Q2. Unit: registered users per 1000 persons (reach) and transactions per registered user per quarter (intensity). Sample: all 35 state units; the other union territories are marked. Source: PhonePe [6]; population from National Commission on Population [4].

5.4 Reach and intensity

Count per person is the product of reach and intensity (Section 3.5). Figure 8 shows the two factors for each state unit in 2026Q2. The horizontal axis gives reach, in registered users per 1000 persons. The vertical axis gives intensity, in transactions per registered user per quarter. The labels are the two-letter codes of Appendix A, and grey labels mark the other union territories.

Reach goes from 238 registered users per 1000 persons in Meghalaya to 943 in Telangana. Intensity goes from 12.2 transactions per user in Manipur to 106.3 in Telangana. With unrounded values, the ratio of the highest to the lowest value is 4.0 for reach and 8.7 for intensity. Thus the units differ more in intensity than in reach. The north-eastern states with the lowest count per person are at the lower left of the figure, with low reach and low intensity.

The two factors go together, but not closely. Their Spearman rank correlation across the main sample is 0.600 in 2026Q2 and 0.309 in 2018Q4. Some units have a much higher rank in one factor than in the other. Goa and Kerala rank 12 places higher in reach than in intensity. Bihar ranks 15 places higher in intensity than in reach.

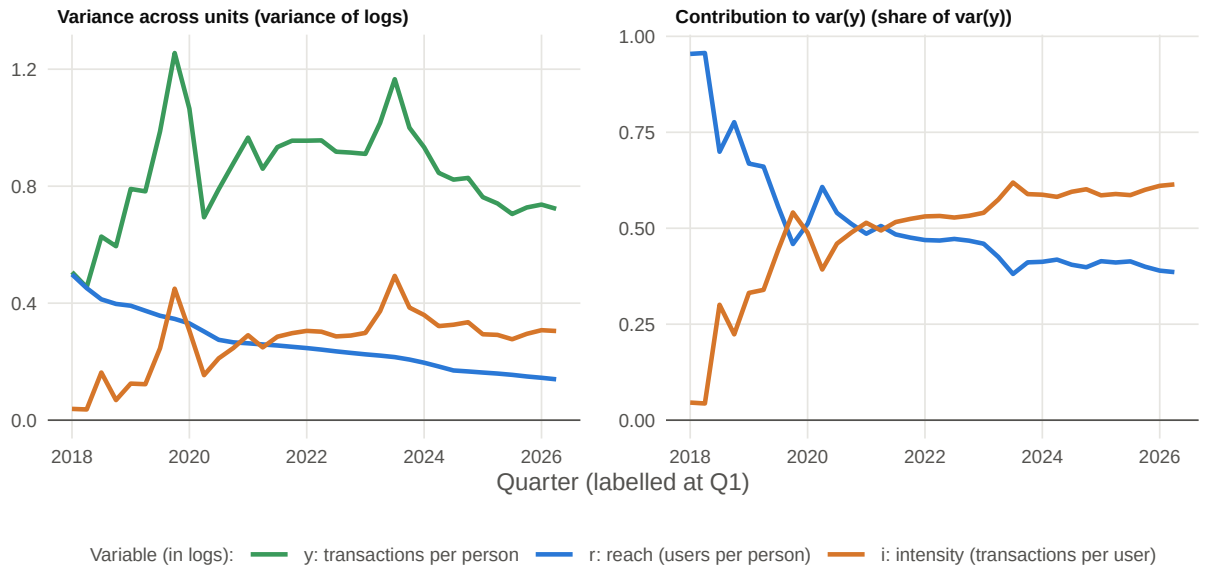


Figure 9: Contributions of reach and intensity to the cross-state variance of the log of transactions per person. Unit: variance of logs (left) and share of var(y) (right). Sample: the 30 units of the main sample. Period: 2018Q1–2026Q2. Left: variances of y, r and i. Right: contributions of reach and intensity to var(y). Source: PhonePe [6]; population from National Commission on Population [4].

Table 6: Variance decomposition of the log of transactions per person: $y = \ln(C/P)$, reach $r = \ln(U/P)$ and intensity $i = \ln(C/U)$. Unit: variance of logs across units; contributions as shares of var(y). Sample: the 30 units of the main sample. Periods: 2018Q4 and 2026Q2. Source: PhonePe [6]; population from National Commission on Population [4].

	2018Q4	2026Q2
n	30	30
var(y)	0.595	0.723
var(r)	0.397	0.140
var(i)	0.069	0.305
$2\,\mathrm{cov}(r, i)$	0.129	0.278
Reach contribution	0.776	0.386
Intensity contribution	0.224	0.614

In the first group, the rank in reach is much higher than the rank in intensity. In the second group, the rank in intensity is much higher than the rank in reach. A cumulative count of registrations can give the first pattern: registrations that are no longer in use increase reach and decrease intensity. The data cannot test this explanation, because they give no count of active users.

5.5 Decomposition of the variance

Section 3.5 divides the cross-state variance of the log of the count per person into two parts. One part goes with reach, and the other part goes with intensity. Figure 9 shows the result for each quarter. The left panel shows the variances of y, r and i across the main sample. The right panel shows the two contributions, which add up to 1 in each quarter. Table 6 gives the values for the two reference quarters.

In 2018Q4, the contribution of reach is 0.776 and the contribution of intensity is 0.224. Thus, in 2018Q4, most of the differences in adoption between units go with differences in reach. In 2026Q2, the contribution of reach is 0.386 and the contribution of intensity is 0.614. Now most

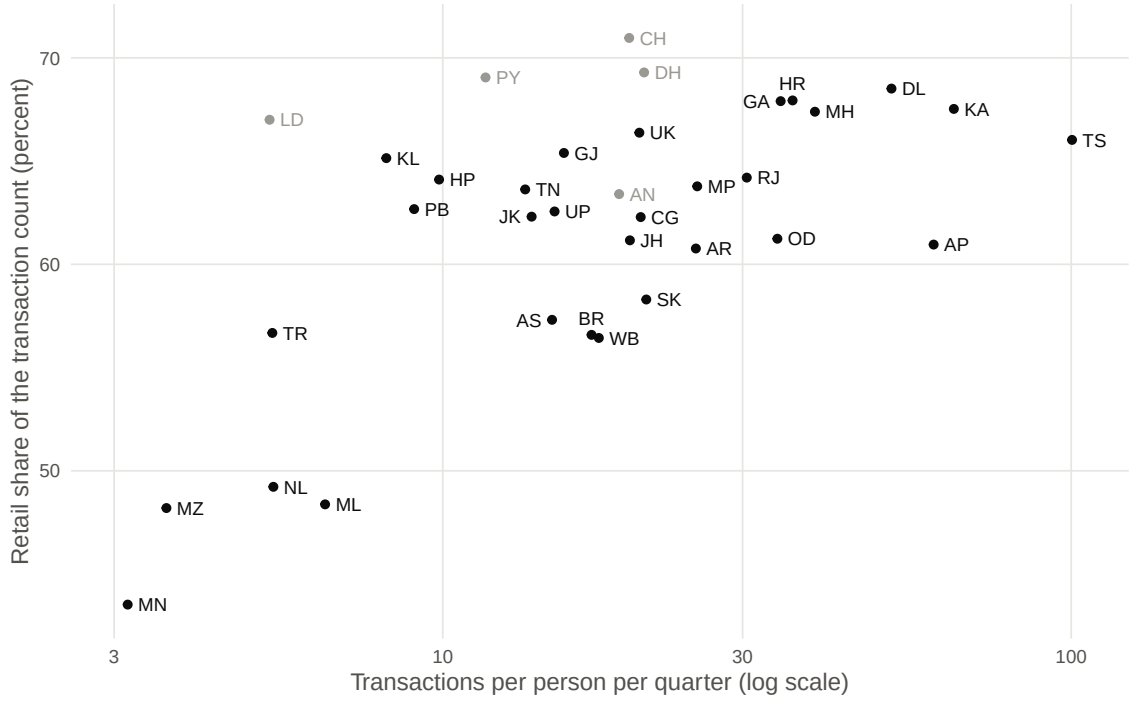


Figure 10: Retail share of the transaction count against transactions per person in 2026Q2. Unit: percent of the state count; transactions per person per quarter. Sample: all 35 state units; the other union territories are marked. Source: PhonePe [6]; population from National Commission on Population [4].

of the differences go with differences in intensity.

The change occurs over several years. In 2018Q1, the contribution of reach is 0.954. Intensity has the larger contribution for the first time in 2019Q4. From 2021Q3, it has the larger contribution in every quarter.

The variances in the left panel show how this change occurs. The variance of r falls from 0.397 in 2018Q4 to 0.140 in 2026Q2. It decreases in every quarter of the period. The variance of i increases from 0.069 to 0.305. Thus the units become more similar in reach and less similar in intensity. The term $2\,\mathrm{cov}(r, i)$ in Table 6 also increases. This agrees with the higher rank correlation of reach and intensity in 2026Q2 (Section 5.4).

The variance of y does not move in one direction. It increases from 0.595 in 2018Q4 to its largest value, 1.255, in 2019Q4. It then decreases. After 2020, its largest value is 1.166, in 2023Q3. In 2026Q2, it is 0.723. The variance of i has its largest value, 0.493, in 2023Q3. Thus the largest value of $\mathrm{var}(y)$ after 2020 goes with a large spread in intensity.

The decomposition describes how the differences between units divide between reach and intensity. It does not show a cause. Part 3 of the project uses the change of the differences over time in its test of convergence. This study gives only the values.

5.6 Retail share

Figure 10 shows the retail share of each state unit against its count per person in 2026Q2. The horizontal axis has a log scale. Grey labels mark the other union territories.

In the main sample, the retail share goes from 43.5 percent in Manipur to 68.5 percent in Delhi. Manipur also has the lowest count per person. The Spearman rank correlation of the retail share and the count per person is 0.625. Thus the units with more transactions per person tend to have a larger share of payments to merchants. Of the other union territories, 3 have a

retail share above the highest value of the main sample. They are outside the main sample, so they do not enter the correlation.

In 2018Q4, the same coefficient is −0.527, with the opposite sign. The type shares of 2018 change by large amounts from quarter to quarter (Section 4.4). Thus the study does not interpret the coefficient of 2018Q4.

The correlation of 2026Q2 does not show a cause. More payments to merchants can increase adoption, or high adoption can bring more merchants to the app. The data of this study cannot separate these two directions.

6 Limitations of the data

The panel measures the use of one app. PhonePe Pulse covers the transactions of PhonePe, not all UPI transactions. The share of PhonePe in UPI can differ between states. If it does, the panel shows differences between states that are not differences in UPI use. The share can also change over time. Then the growth rates of this study differ from the growth of UPI. This study does not measure the share of PhonePe. Part 2 compares the PhonePe values with the NPCI state data for the quarters from April 2025.

The user count is a count of registrations, not of active users or of persons. The count does not fall in any of the 1,155 pairs of consecutive quarters (Section 3.3). This result agrees with a cumulative count. Such a count keeps the users who no longer use the app. Thus reach can increase in a state while the use of the app does not. The data do not show whether one person can have more than one registration. They also do not show whether a registration moves to a new state when a user moves. Thus reach is not the share of persons who use the app.

The repository does not document how PhonePe assigns a transaction or a user to a state. The README gives the hover count as the transactions done in the quarter for the state. It gives the top user lists as the states that users registered from. It does not say whether the state comes from the address of the user, the location of the device or the location of the merchant. Thus the panel cannot show where a payment between two states goes. A state with many persons who live or work in another state can have values that do not match its resident population.

The population values are projections, not counts. NCP 2020 makes them from the 2011 Census and the Sample Registration System (Section 2). The study has no population count for any quarter of the panel. An error in a projection goes directly into each per-person value of that state, and the size of the errors is not known. The per-person values of Dadra and Nagar Haveli and Daman and Diu, with the largest projected growth, depend most on the projection (Section 3.4). For Goa and most union territories, the report uses a mathematical method that does not project fertility, mortality and migration.

All amounts are nominal. The amount per person and the average value thus include the change in prices over the period. A change in the average value can come from a change in prices or from a change in the size of payments. This study cannot separate the two. The count is a number of transactions, not a value. Thus the growth of the count and the count per person need no price adjustment. Part 2 converts the amounts to real values with the consumer price indices [3].

The type shares change by large amounts in 2018. The national retail share moves between 38.4 and 64.2 percent in that year. The data do not show whether these changes come from changes in use or from the restated definitions of the types (Section 2.1). For this reason, the study uses 2018Q4 and not 2018Q1 as the first reference quarter, and it does not interpret the type shares of 2018. The type files also give counts only, so the study cannot divide the amount by type.

7 Conclusions

This part builds a state-quarter panel of PhonePe adoption for 35 state units and 34 quarters, from 2018Q1 to 2026Q2. The denominator is the quarterly population from the NCP 2020 projections. All data checks pass. The type counts sum to the state counts, and the state files sum to the national files in each quarter. The panel gives the findings below. All values are nominal.

First, the national count grows at an annual rate of 82.3 percent from 2018Q4 to 2026Q2. It reaches 38.66 billion transactions in 2026Q2, or 27.11 transactions per person per quarter. The growth becomes lower over time. The median growth per quarter is 17.7 percent up to 2024Q2 and 5.9 percent in the last eight quarters. The count falls only in 2020Q2, with a change of −10.6 percent.

Second, the average transaction value has a peak of Rs. 1,907 in 2020Q3 and falls to Rs. 1,177 in 2026Q2. In the same years, the retail share of the count increases from 37.3 percent in 2021Q2 to 63.9 percent. This pattern agrees with more small payments to merchants. The data do not give the amount by type, so the study does not test this explanation.

Third, the units of the main sample differ in adoption, and their order is stable. The ratio of the highest to the lowest count per person in the main sample is 22.6 in 2018Q4 and 31.8 in 2026Q2. The Spearman rank correlation of the state values in the two quarters is 0.892.

Fourth, five units have most of the count. In 2026Q2, the five units with the largest count have 52.6 percent of the count and 37.5 percent of the population. Half of the national count comes from 5 units.

Fifth, at the end of the period, the differences between units go more with intensity than with reach. The contribution of reach to the cross-state variance of the log count per person is 0.776 in 2018Q4 and 0.386 in 2026Q2. The contribution of intensity increases from 0.224 to 0.614.

Sixth, the units with more transactions per person tend to have larger retail shares. In 2026Q2, the Spearman rank correlation of the retail share and the count per person is 0.625. The data do not show the direction of this relation.

The later parts of the project use this panel. Part 2 converts the amounts to real values with state consumer price indices. It also compares the PhonePe values with the NPCI state data to measure the share of PhonePe in UPI. It shows the measures on maps. Part 3 uses the count per person of the main sample to test convergence. It tests whether the states with low adoption move closer to the leading states. The results of this part are descriptive. They do not answer the questions of Parts 2 and 3.

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A The state units

The names in the last two columns are the raw names after the normalisation of the state crosswalk: lower case, with "and" in place of "&". A semicolon separates the raw names that the crosswalk sums into one unit.

Table 7: The 35 state units of the panel: display name, `state_id`, two-letter code [8], membership of the main sample of 30, and the raw names that map to the unit in the PhonePe Pulse hover file for 2026Q2 [6] and in NCP 2020, Table 8 [4]. Source: state crosswalk (`state_crosswalk.csv`) and state labels (`state_labels.csv`).

Name	state_id	Code	Main	PhonePe raw names	NCP raw names
Andaman and Nicobar Islands	andaman_nicobar	AN	No	andaman and nicobar islands	andaman and nicobar islands
Andhra Pradesh	andhra_pradesh	AP	Yes	andhra pradesh	andhra pradesh
Arunachal Pradesh	arunachal_pradesh	AR	Yes	arunachal pradesh	arunachal pradesh
Assam	assam	AS	Yes	assam	assam
Bihar	bihar	BR	Yes	bihar	bihar
Chandigarh	chandigarh	CH	No	chandigarh	chandigarh
Chhattisgarh	chhattisgarh	CG	Yes	chhattisgarh	chhattisgarh
Dadra and Nagar Haveli and Daman and Diu	dnh_dd	DH	No	dadra and nagar haveli and daman and diu	dadra and nagar haveli; daman and diu
Delhi	delhi	DL	Yes	delhi	nct of delhi
Goa	goa	GA	Yes	goa	goa
Gujarat	gujarat	GJ	Yes	gujarat	gujarat
Haryana	haryana	HR	Yes	haryana	haryana
Himachal Pradesh	himachal_pradesh	HP	Yes	himachal pradesh	himachal pradesh
Jammu and Kashmir (with Ladakh)	jammu_kashmir	JK	Yes	jammu and kashmir; ladakh	jammu and kashmir; ladakh

Name	state_id	Code	Main	PhonePe raw names	NCP raw names
Jharkhand	jharkhand	JH	Yes	jharkhand	jharkhand
Karnataka	karnataka	KA	Yes	karnataka	karnataka
Kerala	kerala	KL	Yes	kerala	kerala
Lakshadweep	lakshadweep	LD	No	lakshadweep	lakshadweep
Madhya Pradesh	madhya_pradesh	MP	Yes	madhya pradesh	madhya pradesh
Maharashtra	maharashtra	MH	Yes	maharashtra	maharashtra
Manipur	manipur	MN	Yes	manipur	manipur
Meghalaya	meghalaya	ML	Yes	meghalaya	meghalaya
Mizoram	mizoram	MZ	Yes	mizoram	mizoram
Nagaland	nagaland	NL	Yes	nagaland	nagaland
Odisha	odisha	OD	Yes	odisha	odisha
Puducherry	puducherry	PY	No	puducherry	puducherry
Punjab	punjab	PB	Yes	punjab	punjab
Rajasthan	rajasthan	RJ	Yes	rajasthan	rajasthan
Sikkim	sikkim	SK	Yes	sikkim	sikkim
Tamil Nadu	tamil_nadu	TN	Yes	tamil nadu	tamil nadu
Telangana	telangana	TS	Yes	telangana	telangana
Tripura	tripura	TR	Yes	tripura	tripura
Uttar Pradesh	uttar_pradesh	UP	Yes	uttar pradesh	uttar pradesh
Uttarakhand	uttarakhand	UK	Yes	uttarakhand	uttarakhand
West Bengal	west_bengal	WB	Yes	west bengal	west bengal

B Reproduction

The commands below run from the repository root. They fetch the raw files, build the interim files, write the figures, tables and numbers, and build this document.

```
make data analysis report
```

The PhonePe Pulse files are at the commit `257bf38e3095020f9f516c4c72c10bdc348781b1`. The study reads these input files:

- `data/interim/phonepe.csv`
- `data/interim/phonepe_types.csv`
- `data/interim/phonepe_checks.csv`
- `data/interim/population.csv`
- `src/R/lib/state_labels.csv`

The script `src/R/part1_study.R` writes the figures to `output/study/part1/figures/`, the tables to `output/study/part1/tables/` and the numbers to `output/study/part1/numbers.tex`. The file `numbers.tex` is generated. Do not edit it. A number in the text comes from this file with the command `\val`, and an unknown key stops the compile.